Lecture 14

Topological quantum field theory: a synthesis

Objectives. *The course has built duals (Lecture 4), F-symbols (Lecture 6), modular data (Lecture 10), algebras (Lecture 11) and centres (Lecture 12). A topological quantum field theory assigns vector spaces to spaces and linear maps to spacetimes between them. In one dimension it is determined by a dual pair. In two dimensions it is determined by a commutative Frobenius algebra. Finite groups give examples called Dijkgraaf–Witten theories. In three dimensions a spherical fusion category gives a Turaev–Viro state sum. For a unitary category its Hamiltonian form is a string net. A modular category gives the Reshetikhin–Turaev theory, related to the state sum through the centre. These constructions cover selected structures of the course. Turaev–Viro theories are non-chiral;* $\mathrm{Rep}(\mathrm{SU}(2))$ *and generic* U_q *are not finite inputs.*

By the end of this lecture you will be able to:

- Define a TQFT as a symmetric monoidal functor, and prove that its state spaces are finite-dimensional, with $Z(\Sigma \times S^1) = \dim Z(\Sigma)$.
- Define Frobenius algebras in pictures, state the classification of 2D TQFTs, and compute partition functions with the handle element.
- Compute 2D Dijkgraaf–Witten partition functions by Mednykh's formula.
- State the Turaev–Viro and Reshetikhin–Turaev constructions and their relation through the centre, with the framing anomaly, and count torus states.

14.1 Atiyah's axioms; 1D

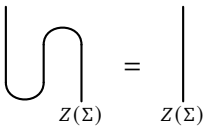
The *bordism category* Bord_n has as objects closed oriented $(n-1)$-manifolds, and as morphisms $\Sigma \to \Sigma'$ the compact oriented n-manifolds M with $\partial M = \bar{\Sigma} \sqcup \Sigma'$, up to diffeomorphism fixing the boundary. Here $\bar{\Sigma}$ is Σ with its orientation reversed. Composition glues along the common boundary. Disjoint union $\sqcup$, with the swap of components, makes this category symmetric monoidal (Kock 1.2–1.3; Carqueville–Runkel §2). The category Cob_2 of Lecture 1 is Bord_2. Time runs upward, as everywhere in the course.

Definition 14.1 (TQFT)**.** An n-dimensional *topological quantum field theory* is a symmetric monoidal functor $Z\colon \mathrm{Bord}_n \to \mathrm{Vect}$ (Kock 1.3.32; Carqueville–Runkel Def. 2.1).

The target is *symmetric*, not braided: exchanging two spaces is the flip of vector spaces. A closed n-manifold is a bordism $\emptyset \to \emptyset$, so its image is a number, the *partition function*.

Theorem 14.2 (Atiyah's axioms, consequences · *sketched; (i)–(ii) proved*)**.** *(i) Each* $Z(\Sigma)$ *is finite-dimensional, with* $Z(\bar{\Sigma}) \cong Z(\Sigma)^*$. *(ii)* $Z(\Sigma \times S^1) = \dim Z(\Sigma)$. *(iii) A one-dimensional TQFT is determined, up to isomorphism, by the finite-dimensional space* $V = Z(\mathrm{pt}_+)$*, and every such* V *occurs (Carqueville–Runkel Thm. 3.2).*

Proof. (i) The cylinder $\Sigma \times [0,1]$, bent, is a bordism $\bar{\Sigma} \sqcup \Sigma \to \emptyset$ and one $\emptyset \to \Sigma \sqcup \bar{\Sigma}$; straightening the bend is a diffeomorphism, so their images ev and coev satisfy the snake equations of Definition 4.1. In one dimension these are pictures of the course's cups and caps:



(14.1)

Write $\mathrm{coev}(1) = \sum_{i=1}^{k} a_i \otimes b_i$, a finite sum, with $a_i \in Z(\Sigma)$ and $b_i \in Z(\bar{\Sigma})$. The snake equation says $v = \sum_i a_i \,\mathrm{ev}(b_i \otimes v)$ for every $v \in Z(\Sigma)$, so $a_1, \dots, a_k$ span $Z(\Sigma)$; and ev identifies $Z(\bar{\Sigma})$ with $Z(\Sigma)^*$. (ii) $\Sigma \times S^1$ is the cylinder with its ends glued, and the gluing passes one copy of Σ past the other: it is $\mathrm{ev} \circ P \circ \mathrm{coev}$, with P the flip of Vect, so $\sum_i \mathrm{ev}(b_i \otimes a_i)$, the trace of the identity, which is $\dim Z(\Sigma)$:



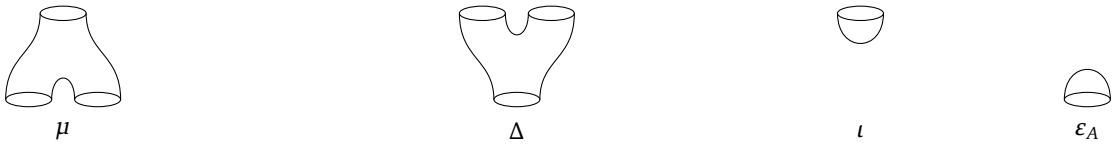
$$= \dim Z(\Sigma). \tag{14.2}$$

(iii) Every 1-manifold with boundary is a composite of the elbows and the identities, so Z is fixed by V, ev and coev; the relations between them are the snake equations, so any dual pair defines a theory. ∎

So a one-dimensional theory is Lecture 4's duality, and the circle is the loop $\dim V$ of Equation (4.7).

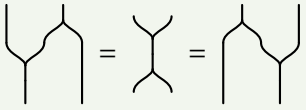
14.2 Frobenius algebras; 2D TQFT is commutative Frobenius

In two dimensions the objects are disjoint unions of circles, and every surface with boundary is glued from a few pieces:



the pair of pants, read upward, merges two circles into one; turned over it splits one into two; the disc with its circle on top creates a circle from nothing; the disc with its circle below annihilates one. (Kock calls the first disc a "cap"; here the words cup and cap stay with Lecture 4.) With $A = Z(S^1)$ they are maps $\mu : A \otimes A \to A$, $\Delta : A \to A \otimes A$, $\iota : \mathbb{C} \to A$ and $\varepsilon_A : A \to \mathbb{C}$.

Definition 14.3 (Frobenius algebra)**.** A *Frobenius algebra* is $(A, \mu, \iota, \Delta, \varepsilon_A)$, an algebra (Definition 11.1) and a coalgebra on the same space (Δ coassociative with counit ε_A: the algebra axioms with the arrows reversed), satisfying the *Frobenius relation*



(14.3)

that is, $(\mathrm{id} \otimes \mu)(\Delta \otimes \mathrm{id}) = \Delta \circ \mu = (\mu \otimes \mathrm{id})(\mathrm{id} \otimes \Delta)$ (Kock Lemma 2.3.19). It is *commutative* if $\mu \circ P = \mu$, and *special* if $\mu \circ \Delta = \mathrm{id}$ (Heunen and Vicary Def. 5.5).

Equivalently, $\varepsilon_A \circ \mu$ is a nondegenerate pairing with $\varepsilon_A(xy \cdot z) = \varepsilon_A(x \cdot yz)$, and then Δ is determined (Kock 2.2.5, Props. 2.3.22 and 2.3.24). For instance $\mathbb{C}[G]$ with $\varepsilon_A(g) = \delta_{g,1}$, or the algebra of $n \times n$ matrices with the trace.

Theorem 14.4 (2D TQFTs · *cited; the bijection proved from the cited presentation*)**.** *$Z \mapsto Z(S^1)$ is a bijection, up to isomorphism, between 2D TQFTs and commutative Frobenius algebras (Abrams; Kock Thm. 3.3.2; Carqueville–Runkel Thm. 3.5, which make it an equivalence of groupoids).*

Proof. Bord_2 is generated, as a symmetric monoidal category, by the four pieces above and the cylinder, subject to relations which say exactly that they form a commutative Frobenius object: associativity and unit, coassociativity and counit, the Frobenius relation, and commutativity, which uses the swap of the two legs of the pants (Kock Prop. 1.4.13 and 1.4.24–1.4.40; Carqueville–Runkel Thm. 3.6) · *cited*. A symmetric monoidal functor therefore sends S^1 to a commutative Frobenius algebra; conversely a commutative Frobenius algebra gives values on the generators that satisfy the relations, hence a functor. ∎

Commutativity comes from the swap of the legs of the pants, which a symmetric Z sends to the flip of Vect; a Z that is only monoidal can give graded-commutative algebras (Kock 3.3.3). Every closed surface is a composite, and the genus-g surface is

$$Z(\Sigma_g) = \varepsilon_A\big(w^g\big), \qquad w = \mu \circ \Delta \circ \iota = \tag{14.4}$$



with the *handle element* w: each handle is a split followed by a merge.

Example 14.5 (Semisimple theories)**.** Let $A = \mathbb{C}^n$, with orthogonal idempotents $e_1, \dots, e_n$ and $\varepsilon_A(e_i) = \lambda_i \neq 0$. The pairing $\varepsilon_A(e_i e_j) = \delta_{ij}\lambda_i$ is nondegenerate, so this is a commutative Frobenius algebra, with $\Delta(e_i) = \lambda_i^{-1} e_i \otimes e_i$ and $w = \sum_i \lambda_i^{-1} e_i$. Hence

$$Z(\Sigma_g) = \sum_i \lambda_i^{1-g},$$

so $Z(S^2) = \sum_i \lambda_i$ and $Z(T^2) = n = \dim A$, as Theorem 14.2 requires. Each idempotent alone gives a theory with one-dimensional state spaces and $Z(\Sigma_g) = \lambda_i^{1-g}$, and on connected surfaces Z is their sum. Every commutative semisimple algebra over $\mathbb{C}$ is some $\mathbb{C}^n$, so this is the whole story for semisimple A, the case of the next section.

14.3 Dijkgraaf–Witten

Theorem 14.6 (2D Dijkgraaf–Witten · *cited; the character sum sketched*)**.** *For a finite group G, the class functions on G with the convolution $(f * h)(x) = \sum_{yz=x} f(y)h(z)$ and $\varepsilon_A(f) = f(1)/|G|$ form a commutative Frobenius algebra, and its TQFT has*

$$\begin{aligned} Z(\Sigma_g) &= \frac{\#\operatorname{Hom}(\pi_1\Sigma_g, G)}{|G|} \\ &= \sum_{\chi} \left(\frac{|G|}{\dim \chi}\right)^{2g-2}, \end{aligned}$$

the sum over the irreducible characters (Mednykh; Snyder eq. (1); Freed Ex. 4.17).

Sketch. Character orthogonality (*Symmetries*, Lecture 5) gives the central idempotents as class functions:

$$e_\chi(x) = \frac{\dim \chi}{|G|}\,\chi(x^{-1}) \qquad (x \in G).$$

They satisfy $e_\chi * e_{\chi'} = \delta_{\chi\chi'} e_\chi$. Since $\chi(1) = \dim \chi$, the counit is

$$\varepsilon_A(e_\chi) = \frac{e_\chi(1)}{|G|} = \left(\frac{\dim \chi}{|G|}\right)^2.$$

In this basis $\Delta(e_\chi) = \varepsilon_A(e_\chi)^{-1} e_\chi \otimes e_\chi$, so $w = \sum_\chi \varepsilon_A(e_\chi)^{-1} e_\chi$ and $Z(\Sigma_g) = \sum_\chi \varepsilon_A(e_\chi)^{1-g}$. The first expression is Mednykh's theorem (cited; Sheet 14 proves it): it counts flat G-connections, weighted by $1/|G|$ for the gauge symmetry. ∎

For $G = S_3$, with $\dim \chi = 1, 1, 2$:

$$Z(\Sigma_2) = 6^2 + 6^2 + 3^2 = 81,$$

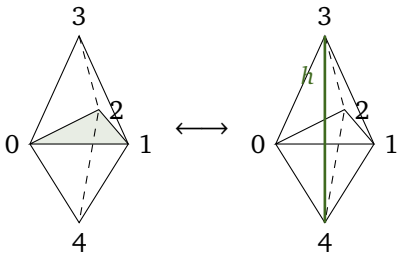
and $\#\operatorname{Hom}(\pi_1 \Sigma_2, S_3) = 486 = 81 \cdot 6$; at genus 1 the value is 3, the number of conjugacy classes (the script counts both). Scaling ε_A by a nonzero constant multiplies $Z(\Sigma_g)$ by that constant to the power $1 - g$. Thus the counit extracting $f(1)$ instead of $f(1)/|G|$ gives a different answer.

14.4 Turaev–Viro and string nets

In three dimensions the input is a spherical fusion category C, and the F-symbols of Lecture 6 become the weights of a state sum.

Theorem 14.7 (Turaev and Viro · *sketched*)**.** *Let C be a spherical fusion category with $\dim C \neq 0$, multiplicity-free as in Lecture 6. Colour the edges of a triangulation of a closed 3-manifold M by simple objects, weight each tetrahedron by a normalised F-symbol and each edge by its dimension, and sum over colourings with a factor $(\dim C)^{-1}$ for each vertex. (In general one also sums over bases of the fusion spaces on the faces, Turaev–Virelizier pp. 273–275.) The result $Z_{\mathrm{TV}}(M)$ does not depend on the triangulation, and extends to a 3D TQFT (Turaev–Virelizier Thms. 13.1 and 13.5; Simon (23.1)).*

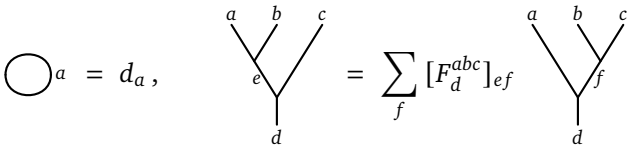
Sketch. Any two triangulations are related by the Pachner moves. The 2–3 move fills the same triangular bipyramid in two ways:



The two tetrahedra share face 012. The three tetrahedra share the new edge 34. Boundary edge labels are fixed. Summing the label h of the new edge gives the intermediate-channel sum in the pentagon (Theorem 6.3), so the two sets of weights agree. The 1–4 move is absorbed by the vertex factor, by the orthogonality of the F-symbols (Simon p. 319). No braiding is used. ∎

For unitary C its Hamiltonian form is the *string net* of Levin and Wen. Put a label of C on each edge of a trivalent lattice in a surface; the ground states are the superpositions of labelled graphs invariant under local moves, the graphical calculus of Lectures 5 and 6: a

loop is d_a, a bubble is removed, and the F-move of Equation (6.1) relates the two ways of splitting d into a, b, c,



$$\bigcirc^{a} = d_a\,, \qquad = \sum_f [F_d^{abc}]_{ef} \tag{14.5}$$

(Levin and Wen, (4)–(7)). The Hamiltonian $H = -\sum_v Q_v - \sum_p B_p$ has commuting projectors as terms: Q_v asks that the three labels at a vertex fuse to 1, and B_p inserts the loop $\sum_s (d_s/\dim C)\, s$ inside a plaquette and fuses it into the edges with F-moves. It projects onto the subspace in which the plaquette can be filled without a puncture (Kirillov Thm. 3.9). Its coefficients are those of Lecture 10's killing loop, but the action is different: a loop is added to the lattice and resolved by F-moves. No braiding of the input C is needed. The strand-encircling projector in Lecture 10 instead uses a braiding and non-degeneracy. The ground space on a surface Σ is $Z_{\mathrm{TV}}(\Sigma)$ (Kirillov Thm. 5.1). For $C = \mathrm{Vec}_{\mathbb{Z}_2}$ it is the toric code, with four ground states on the torus (*Quantum Computing* Thm. 14.2).

14.5 RT and $\mathrm{TV} = \mathrm{RT} \circ \mathcal{Z}$; the outlook

Theorem 14.8 (Reshetikhin and Turaev; Turaev and Virelizier · *cited*). *(i) A modular category C with $p_+ = p_-$ gives a 3D TQFT RT_C; if $p_+ \neq p_-$ the gluing holds only up to a phase, the* framing anomaly *(Bakalov–Kirillov Thm. 4.4.3 and Rem. 4.4.5). (ii) For a spherical fusion category with $\dim C \neq 0$, $Z_{\mathrm{TV},C} \cong \mathrm{RT}_{\mathcal{Z}(C)}$ (Turaev–Virelizier Thm. 17.1, Cor. 17.5). (iii) $\dim \mathrm{RT}_C(T^2)$ is the number of simple objects of C (Bakalov–Kirillov Ex. 4.5.3).*

For a unitary C, $p_- = \overline{p_+}$, so $p_+/p_- = e^{2\pi i c/4}$ by Theorem 10.9: the anomaly vanishes exactly when $c \equiv 0 \pmod 4$, if the square root of $\dim C$ that normalises RT may be taken negative, as Bakalov and Kirillov allow. With the positive root, as unitarity asks, the gluing phase $p_+/\sqrt{\dim C} = e^{2\pi i c/8}$ is 1 exactly when $c \equiv 0 \pmod 8$. A centre has $c = 0$ (Lecture 12), so $\mathrm{RT}_{\mathcal{Z}(C)}$ is anomaly-free, as (ii) needs. Fibonacci has $c = \frac{14}{5}$: $\mathrm{RT}_{\mathrm{Fib}}$ is anomalous, and Fibonacci is not a centre, so no Turaev–Viro theory produces it alone; the closest is $Z_{\mathrm{TV,Fib}} = \mathrm{RT}_{\mathrm{Fib}\boxtimes\mathrm{Fib}^{\mathrm{rev}}}$, the doubled Fibonacci of Levin and Wen (Turaev–Virelizier Cor. 17.5 with Theorem 12.10). By (iii), the torus carries rank $\mathcal{Z}(C)$ states: 4 for $\mathrm{Vec}_{\mathbb{Z}_2}$, the toric code's degeneracy of *Quantum Computing*, and 8 for Vec_{S_3}, where 2D Dijkgraaf–Witten gave $Z(T^2) = 3$ (the script checks both).

For further study, a *fully extended* TQFT assigns data in every lower dimension, down to points. Framed theories also retain a continuously chosen frame of tangent vectors. The cobordism hypothesis classifies such theories by a higher-dimensional duality condition called *full dualizability* (Lurie, proof outline, Thm. 1.2.16). Fusion categories over $\mathbb{C}$ satisfy this condition and give framed 3D theories with a fusion category assigned to a point (Douglas, Schommer-Pries and Snyder, Thm. 1). No result here uses this classification.

Summary.

- *TQFTs:* symmetric monoidal functors $\mathrm{Bord}_n \to \mathrm{Vect}$; state spaces are finite-dimensional, $Z(\Sigma \times S^1) = \dim Z(\Sigma)$, and 1D theories are dual pairs (Definition 14.1 and Theorem 14.2).
- *2D:* commutative Frobenius algebras (Theorem 14.4); $Z(\Sigma_g) = \varepsilon_A(\omega^g)$ with the handle element.

- *Dijkgraaf–Witten:* $Z(\Sigma_g) = \sum_\chi (|G|/\dim\chi)^{2g-2}$; 81 for S_3 at genus 2 (Theorem 14.6).
- *3D:* Turaev–Viro from spherical fusion C, as string nets for unitary C; RT from modular C, anomalous unless $c \equiv 0 \pmod 4$, or modulo 8 with the positive root of $\dim C$; $Z_{\mathrm{TV},C} = \mathrm{RT}_{\mathcal{Z}(C)}$, and the torus counts rank $\mathcal{Z}(C)$ (Theorems 14.7 and 14.8).

References

Kock, *Frobenius Algebras and 2D Topological Quantum Field Theories*: TQFTs, 1.2.23–1.3.32, pp. 30–54; the presentation of 2Cob, Prop. 1.4.13 and 1.4.24–1.4.40, pp. 62–77; Frobenius algebras, 2.2.5, p. 95, Lemma 2.3.19 and Props. 2.3.22 and 2.3.24, pp. 117–119; the classification, Thm. 3.3.2 and 3.3.3, pp. 173–175. Carqueville and Runkel, *Introductory lectures on topological quantum field theory*: Def. 2.1, p. 8; Prop. 2.6, p. 13; Thms. 3.2, 3.5 and 3.6, pp. 21–28. Heunen and Vicary, *Categories for Quantum Theory*, Def. 5.5, p. 150. Snyder, math/0703073, eq. (1), p. 1. Freed, *Lectures on Field Theory and Topology*, Ex. 4.17, p. 59. Turaev and Virelizier, *Monoidal Categories and Topological Field Theory*: multiplicities, pp. 273–275; Thms. 13.1 and 13.5, pp. 275–286; Thm. 17.1, p. 432, and Cor. 17.5, p. 441. Bakalov and Kirillov: Thm. 4.4.3, p. 83, Rem. 4.4.5, p. 87, and Ex. 4.5.3, p. 89. Simon, *Topological Quantum*: state sums, (23.1), p. 317, and p. 319. Levin and Wen, cond-mat/0404617, (4)–(7), p. 5. Kirillov, arXiv:1106.6033, Thm. 3.9, p. 9, and Thm. 5.1, p. 11. Lurie, arXiv:0905.0465, Thm. 1.2.16, p. 14. Douglas, Schommer-Pries and Snyder, arXiv:1312.7188, Thm. 1, p. 4.